

CLIENT SIDE SCRIPTING (22519)

Practical No. 12: Develop a web page for validation of form fields using regular expressions.

Roll No.: 220447

- 1) Write a JavaScript program to validate Name, Age, Date of Birth, Email Id and Password using regular expression.

PERSONAL DETAILS

ACCOUNT

Password length must be atleast 8 characters and maximum 16
Password must contain atleast one UPPERCASE character
Password must contain atleast one number
Password must contain atleast one special character

```
> document.personalDetails.password.value  
< 'A12@q'
```

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport"  
content="width=device-width,  
initial-scale=1.0">  
  <title>Window Object</title>  
  <link rel="stylesheet"  
href="style.css">  
</head>  
<body>  
  <form name="personalDetails"  
action="" id="personalDetails">  
    <div class="personal">  
      <h2>PERSONAL  
DETAILS</h2>  
      <input type="text"  
placeholder="Enter your Name"  
class="max-input"  
onchange="checkName()"  
name="username">
```

```
      <input type="number"  
placeholder="Enter your Age"  
name="age" onchange="checkAge()">  
      <input type="text"  
placeholder="Date of Birth"  
onclick="alertFormat()" name="dob"  
onchange="checkDate()">  
    </div>  
    <div class="social">  
      <h2>ACCOUNT</h2>  
      <input type="text"  
placeholder="Enter your Email"  
class="max-input" name="email"  
onchange="checkEmail()">  
      <div class="pw">  
        <input  
type="password"  
placeholder="Password" class="max-  
input" oninput="checkPassword()"  
name="password">  
        <ul  
class="password-rules">
```

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```

        <li
id="rule1">Password length must be
atleast 8 characters and maximum
16</li>
        <li
id="rule2">Password must contain
atleast one UPPERCASE
character</li>
        <li
id="rule3">Password must contain
atleast one number</li>
        <li
id="rule4">Password must contain
atleast one special character</li>
    </ul>
</div>

</div>
<input type="submit"
onclick="return validate()"
class="btn">
</form>
<script>
    var flag = {
        name:false, age:false,
dob:false, email:false,
password:false
    };
    var list =
document.getElementsByClassName('pa
ssword-rules');
    var li = new Array(4);
    var pw =
document.personalDetails.password;
    var pwValidity = 0;
    var alertOnce = true;
    for(let i=1;i<5;i++){
        li[i-1] =
document.getElementById(`rule${i}`)
;
    }
    function
changeColor(element,color){
        element.style.color = color;
    }
    function alertFormat(){
        if(alertOnce){
            alert('Format: DD-MM-
YYYY');
            alertOnce = false;
        }
    }

```

```

function checkPassword(){
    var pwRegex1 = /[A-Z]+/;
    var pwRegex2 = /[0-9]+/;
    var pwRegex3 = /^[A-Za-z0-
9]+/;
    var pwRegex4 = /[a-z]+/;
    if(pwRegex1.test(pw.value)){
        changeColor(li[1],"green");
        pwValidity++;
    }else{
        changeColor(li[1],"red");
        pwValidity--;
    }
    if(pwRegex2.test(pw.value)){
        changeColor(li[2],"green");
        pwValidity++;
    }else{
        changeColor(li[2],"red");
        pwValidity--;
    }
    if(pwRegex3.test(pw.value)){
        changeColor(li[3],"green");
        pwValidity++;
    }else{
        changeColor(li[3],"red");
        pwValidity--;
    }
    if(pwRegex4.test(pw.value)){
        if(pw.value.length >=8
&& pw.value.length<=16){
            changeColor(li[0],"green");
            pwValidity++;
        }else{
            changeColor(li[0],"red");
            pwValidity--;
        }
    }
    if(pwValidity == 4){
        flag.password = true;
    }else{
        flag.password = false;
    }
}
function checkName(){

```

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```
        var regex = /^[A-Za-z
]{3,26}$/;
        var name =
document.personalDetails.username;
        if(!regex.test(name.value)){
            alert("Name length
range : 3-26\nNumbers & Special
character not allowed in name");

        name.classList.add('invalid-
value');
            flag.name = false;
        }else{

        name.classList.remove('invalid-
value');
            flag.name = true;
        }
        function checkAge(){
            if(this.value<12){
                alert('Age not
valid');
                flag.age=false;
            }
            else{
                flag.age=true;
            }
        }
        function checkDate(){
            var regex = /^[0][1-
9]|[12][0-9]|[3][01)][-](0[1-
9]|1[012])[-]([12][0-9][0-9][0-
9])$/;
            var dob =
document.personalDetails.dob;
```

```
        if(regex.test(dob.value)){

            dob.classList.remove('invalid-
value');
                flag.dob=true;
            }
            else{

            dob.classList.add('invalid-
value');
                flag.dob=false;
            }
        }
        function checkEmail(){
            var email =
document.personalDetails.email;
            var regex = /^[a-z0-
9.]+)((gmail|yahoo|hotmail)\.([a-
z]{2,5}))$/;
            if(regex.test(email.value)){
                flag.email = true;

            email.classList.remove('invalid-
value');
            }
            else{
                flag.email=false;

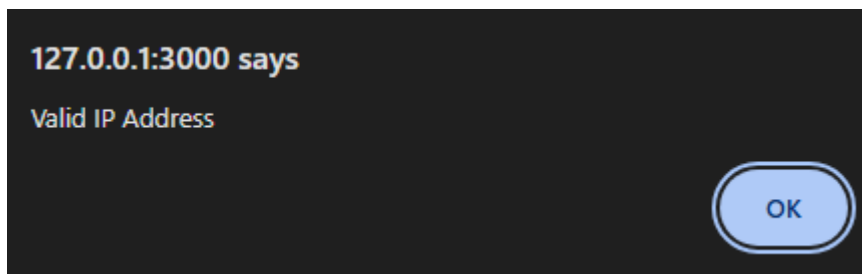
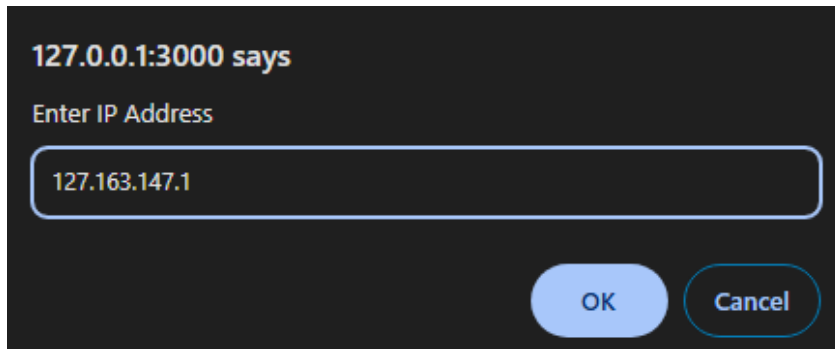
            email.classList.add('invalid-
value');
            }
        }
    }
</script>
</body>
</html>
```

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2) Write a JavaScript program to validate the IP Address using regular expression.



```
<script>
```

```
    var ip = prompt("Enter IP Address");  
    var regex = /^(25[0-5]|2[0-4][0-9]|1[0-9]{2}|[1-9]?[0-9])(\.(25[0-5]|2[0-4][0-9]|1[0-9]{2}|[1-9]?[0-9])){3}$/;  
    if (regex.test(ip)) {  
        alert("Valid IP Address");  
    } else {  
        alert("Invalid IP Address");  
    }  
}
```

```
</script>
```